INCH-POUND

MIL-C-63261 (AR)
AMENDMENT 1
20 January 1990

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TNT, HE, M760
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-63261 (AR) dated 9 October 1980 and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: Standards Military

Add: "MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification"

Add: "MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective"

Delete: "MIL-STD-1235 - Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes"

2.1 Drawings:

9211610-2 and 9278812-2 Delete in their entirety.

2.1 Drawings: 9289185, delete title:

"Cartridge, 105MM, HE, M760, with M200 Propelling Charge Assembly for Howitzer, M204, Loading" and substitute the following:

"Cartridge, 105MM, HE, M760, for Howitzer M119 Loading Assembly"

2.1 Drawings: Add the following:

"12551889 - Projectile, 105MM, HE, M760 Loading Assembly"

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2.1 Product and Package Drawings:

Delete the following in their entirety:

"7549072", "7549073", "9275180", and "9275181".

2.1.1 Add the following paragraph:

"2.1.1 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets of MS standards), the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

3.3.1 Add the following sentence to the end of Table I:

"An Annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast."

3.3.2, Last line, delete: "9211610-2 or dwg. 9278812-2" and substitute "12551889".

Add 3.3.6 and 3.3.7 as follows:

"3.3.6 Metal parts defects. There shall be no cracks, pipes or thin wall sections in the projectile.

3.3.7 Foreign material. There shall be no foreign material in the explosive charge."

3.4.3.2.1, Title, delete the following "(applicable to M572 Fuze only)"

PAGE 5

3.4.3.3, Line 2 and 3 delete:

"shall not be more than 2195 feet per second nor less than 2145 feet per second" and substitute the following:

"shall not be more than 2102 feet per second nor less than 2052 feet per second".

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3.4.3.4 Delete in its entirety and substitute the following:

"3.4.3.4 Pressure. The pressure of the cartridge, fired at +145 degrees Fahrenheit, shall not produce a PIMP (Permissible Individual Maximum Pressure) greater than 57,000 psi when fired in a M119 Howitzer."

3.5 Lines 3, 4, 5 and 6: Delete, "US Army Armament . . . . projectile. A substitute item . . . . . Dover, New Jersey 07801." and substitute the following:

"US Army Armament Research, Development and Engineering Center, for use in radiographing this projectile. A substitute item shall not be used before prior testing and approval by the US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QA, Picatinny Arsenal, New Jersey, 07806-5000."

4.2 Delete and substitute:

"4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection"

Add 4.2.1, 4.2.1.1, 4.2.1.2 and 4.2.1.3

"4.2.1 First article inspection.

4.2.1.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.2.1.2. The first article sample shall consist of:

Part Description	Drawing	Quantity
Cartridge, 105MM, HE, M760, LAP	9289185	52
Projectile, 105MM, HE, M760, LAP	12551889	10

4.2.1.2 Inspection to be performed. See MIL-A-48078 and Table II specified herein.

4.2.1.3 Rejection. See MIL-A-48078."

Add Table II First article as included page 6a.

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TABLE II. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, 105MM, HE, M760 Loading Assembling Packing	1 OF 1	9289185 or 12551889 as applicable
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cartridge, 105MM, HE, M760 Loading, Assembling and Packing (Dwg. 9289185)			
	Examination for defects	52(a)	3.2	4.3.2.10
	Proving Ground Test (Muzzle Velocity-Functioning)	30(a)	3.4	4.3.3.5.4, 4.4.2
	Proving Ground Test (Safety)	20(a)	3.4	4.3.3.5.2, 4.4.2
	Examination for defects	b(a)	3.2	4.3.2.6 thru 4.3.2.9
	Projectile, 105MM, HE, M760 Loading Assembly (Dwg. 12551889)			
	Examination for defects	10	3.2	4.3.2.3 thru 4.3.2.4
	Radiographic Inspection	10	3.3	4.4.1

4 (6a)

NOTES: (a) Use same samples for these tests and examinations.
(b) Two cartridges shall be disassembled to allow checking of the examination from para. 4.3.2.6 thru 4.3.2.9.

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4.3.2.1 Block, Drawing Number: delete: "9211610-2 of 9278812-2 as applicable" and substitute "12551889"

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4.3.2.2 Block, Drawing Number: delete: "9211610-2 or 9278812-2 as applicable" and substitute "12551889".

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4.3.2.3 Block, Drawing Number: delete: "9211610-2" and substitute "12551889".

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4.3.2.4 Block, Drawing Number: delete: "9211610-2" and substitute "12551889".

PAGES 12 and 13

4.3.2.5 Delete in its entirety.

PAGE 19

4.3.2.11 Block, Drawing Number: delete: "7549073 or"

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4.3.2.12 Block, Drawing Number: delete "7549073 or"

PAGE 21

4.3.2.13 Block, Drawing Number: delete in its entirety: "7549072, 9258027, 9271850 or 9275181 as applicable" and substitute the following: "9258027 or 9271850 as applicable".

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4.3.3.2 Delete the last para in its entirety: "The sample plan . . . . . equipment needed for this test is CD." and substitute the following:

"This test shall be performed on all loaded projectile assemblies (100 percent inspection). The radiographic test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4. Any assembly which has any one of the above critical defects shall be classed defective and removed from the lot."

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Add paragraphs: 4.3.3.2.1 and 4.3.3.2.2 as follows:

"4.3.3.2.1 Metal parts defects (Critical defect). All radiographs of projectile section loading assemblies shall be examined for evidence of cracks, pipes and thin wall sections in the warhead. The presence of any such indication shall cause that assembly to be classed defective and removed from the lot. (See Section 6.18).

4.3.3.2.2 Foreign material (Critical defect). All radiographs of projectile section loading assemblies shall be examined for foreign material in the explosive charge. The presence of any foreign material shall cause that assembly to be classed defective and removed from the lot."

4.3.3.3 Paragraph title, delete: "9211610 or 9278812 as applicable," and substitute "12551889".

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4.3.3.5.2.1; 4.3.3.5.2.2; 4.3.3.5.3 and 4.3.3.5.4 delete: "M572 Fuze" wherever it appears (5 places) and substitute M739 Fuze".

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4.3.3.5.5 and 4.3.3.5.5.1 delete in their entirety and substitute the following:

"4.3.3.5.5 Pressure test, with fuze, M739 (see 3.4.3.4)-Major defect. for the first three lots manufactured, the cartridges used in 4.3.3.5.2.1 (conditioned to +145 degrees Fahrenheit) shall be used for this test. If any individual pressure exceeds requirement specified in 3.4.3.4 when tested as specified in 4.4.2.2.1, the lot shall be rejected.

4.3.3.5.5.1 Pressure test, with fuze, M739 (see 3.4.3.4)-Major defect. After three (3) consecutive lots have complied with the criteria of 4.3.3.5.5, the cartridges used in 4.3.3.5.2.2 (conditioned to +145 degrees Fahrenheit) shall be used for this test. If any individual pressure exceeds requirement specified in 3.4.3.4 when tested as specified in 4.4.2.2.1, the lot shall be rejected."

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4.3.4 Delete in its entirety and substitute the following:

"4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6 herein."

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Add paragraph 4.4.1.3 as follows:

"4.4.1.3 Operator qualifications. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

4.4.2, Line 4, delete: "M204 howitzer" and substitute "M119 howitzer".

4.4.2.1, Paragraph title: delete "M572 Fuze" and substitute "M739 Fuze".

4.4.2.1 Change "informational control projectiles/rounds/ samples" to "M760 master calibration cartridges" wherever it appears (4 places).

4.4.2.1, Delete "1146 Mils" and "381 Mils" wherever they appear (2 places) and substitute "750 Mils ± 50 Mils" and "380 Mils ± 50 Mils" respectively.

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4.4.2.2, Line 2, Add the following sentence after the first sentence:

"Each group shall be temperature conditioned for 16 hours minimum."

4.4.2.2, Line 10, delete: "1146 Mils" and substitute "750 Mils ± 50 Mils".

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4.4.2.2, Line 11, delete the last sentence and substitute the following:

"Pressure data for group 2 shall be recorded for information only."

Add paragraph 4.4.2.2.1:

"4.4.2.2.1 PIMP pressure (+145 degrees Fahrenheit). The projectile samples and test conditions for this test shall be group 1 of paragraph 4.4.2.2. The pressure data for each round shall be recorded."

4.4.2.3 Delete "and pressure" from paragraph title.

4.4.2.3, Line 5, delete: "1146 Mils" and substitute "750 Mils ± 50 Mils".

5., 5.1, 5.1.1.1, 5.2, 5.2.1, 5.2.1.1, 5.3 and 5.3.1 delete in their entirety and substitute the following:

"5. PACKING

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Fiber container. The cartridges shall be preserved and packaged in accordance with drawing 9258026.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Outer container. The fiber container shall be packed in accordance with either drawing 9271850 or drawing 9258027.

5.3 Marking.

5.3.1 Marking. Marking of the fiber container shall be in accordance with drawing 9258026. Marking of the outer box shall be in accordance with either drawing 9271850 or drawing 9258027."

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6.1, Line 2, delete: "Howitzer, M2Ø4" and substitute "Howitzer, M119".

6.2.1 Add new paragraph:

"6.2.1 Data card. Data cards are required and shall be prepared. See MIL-A-48Ø78 (AR)."

6.3 Delete in its entirety and substitute the following:

"6.3 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC."

Delete 6.7, 6.7.1, 6.7.2 and 6.7.3 in their entirety and substitute the following:

6.7 Submission of inspection designs for approval. See MIL-A-48Ø78. Submit equipment designs as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, New Jersey Ø78Ø6-5ØØØ.

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6.8 Lines 3 and 4: delete the following:

"Army Armament Research and Development Command, Dover, New Jersey Ø78Ø1, ATTN: DRDAR-QAR-Q" and substitute the following:

"Army Armament Research, Development and Engineering Center, Picatinny Arsenal, New Jersey Ø78Ø6-5ØØØ, ATTN: AMSMC-QAR-Q (D)".

6.1Ø, Lines 5, 6, 7 and 8: delete the following:

"Commander
U.S. Army Armament Research and Development Command
ATTN: DRDAR-QAR-Q and DRDAR-LCA-G
Dover, New Jersey Ø78Ø1" and substitute the following:

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"Commander
U.S. Army Armament Research, Development and Engineering Center
Picatinny Arsenal, New Jersey 07806-5000
ATTN: AMSMC-QAR-Q (D) and SMCAR-FSA-P"

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6.15, Lines 4 and 5 delete the following:

"Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801" and substitute the following:

"Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-Q (D), Picatinny Arsenal, New Jersey 07806-5000".

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6.16, delete: "M752 Fuze" wherever it appears (4 places) and substitute "M739 Fuze".

6.16, delete: "1146 Mils" wherever it appears (6 places) and substitute "750 Mils ± 50 Mils".

6.16 Test 1. Change title from "Safety" to "Safety and Pressure".

Test 3. Change title from "Informational Control Rounds" to "M760 Master Calibration Cartridges".

Test 5. Change title from "Muzzle Velocity and Pressure" to "Muzzle Velocity".

6.16, delete: "381 Mils" wherever it appears (2 places) and substitute "380 Mils ± 50 Mils".

Add paragraph 6.18 as follows:

"6.18 Rejected projectile metal parts. The procuring activity will be notified of any projectile rejected for metal parts defects found upon inspection of the radiographs. The rejected projectile will be held pending disposition by the procuring activity."

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-A746)

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